

Assignment 1 - 02429 Analysis of correlated data: Linear mixed models

Group 4

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Exercise 1

Visualisation

Exploring the data, Figure 1 shows the percentage of conception by bull, displayed in a box plot. Figure 1 shows a large variance in the samples from bull 4, while others, e.g., bull 3 have a lower variance. Furthermore, bull 5 has a high percentage of conception compared to the rest of the sample.

Estimation of percentage of conception

Considering the bulls as a random sample of a population, and assuming that observations from each bull are correlated, the percentage of conception can be estimated with a random-effects model with bull as a random factor.

Model

$$y_{ij} = \mu + a_i + \epsilon_{ij} \quad (1)$$

Where:

- y_{ij} represents the j -th percentage measurement for the i -th bull.
- μ is the overall true average percentage of conceptions across the entire population.
- a_i is the random effect for the i -th bull, assumed to be normally distributed with a mean of zero and a variance of σ_a^2 ($a_i \sim N(0, \sigma_a^2)$).
- ϵ_{ij} is the residual measurement error for each specific sample, assumed to be normally distributed with a mean of zero and a variance of σ_ϵ^2 ($\epsilon_{ij} \sim N(0, \sigma_\epsilon^2)$).

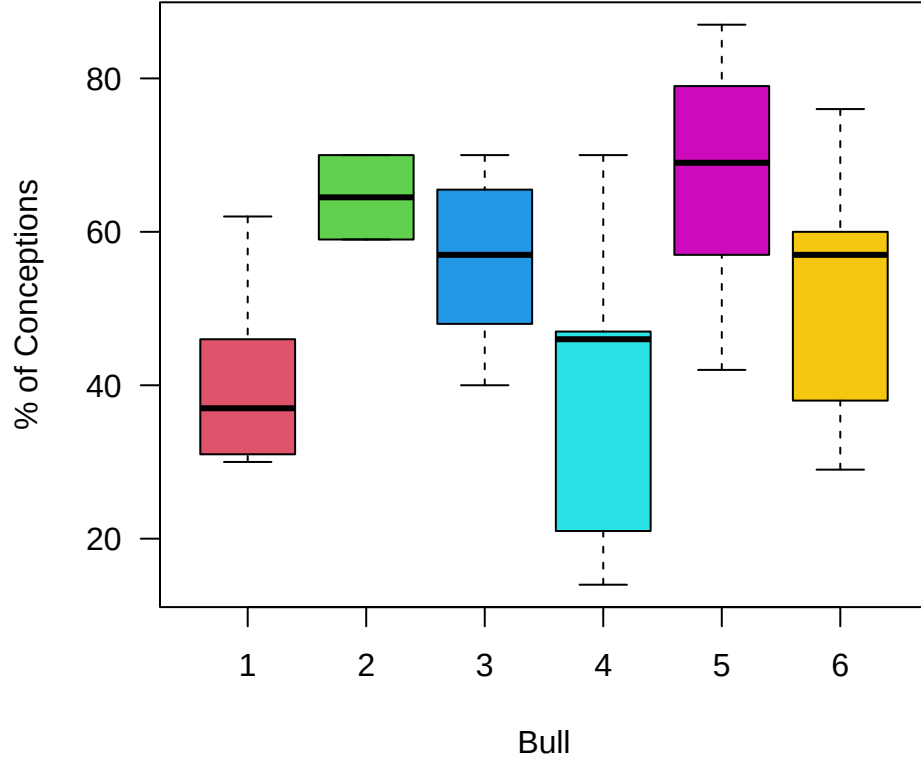


Figure 1: Percentage of conception of bulls.

Results

$\mu = 53.32\%$, 95% CI [43.68%, 63.05%]

Bull-to-bull variability ($\hat{\sigma}_a^2$): 76.82

Within-bull/Residual variability ($\hat{\sigma}_\epsilon^2$): 248.7

To calculate the heritability parameter, h , Equation 2 is used.

$$h = \frac{4\sigma_a^2}{\sigma_a^2 + \sigma_\epsilon^2} \quad (2)$$

$h = 94.39\%$

Exercise 2

Visualisation

Figure 2 shows boxplots of breaking strength by site (Figure 2a), and by treatment (Figure 2b). Figure 2b could indicate that the treatment does not increase breaking strength, or even decreases it.

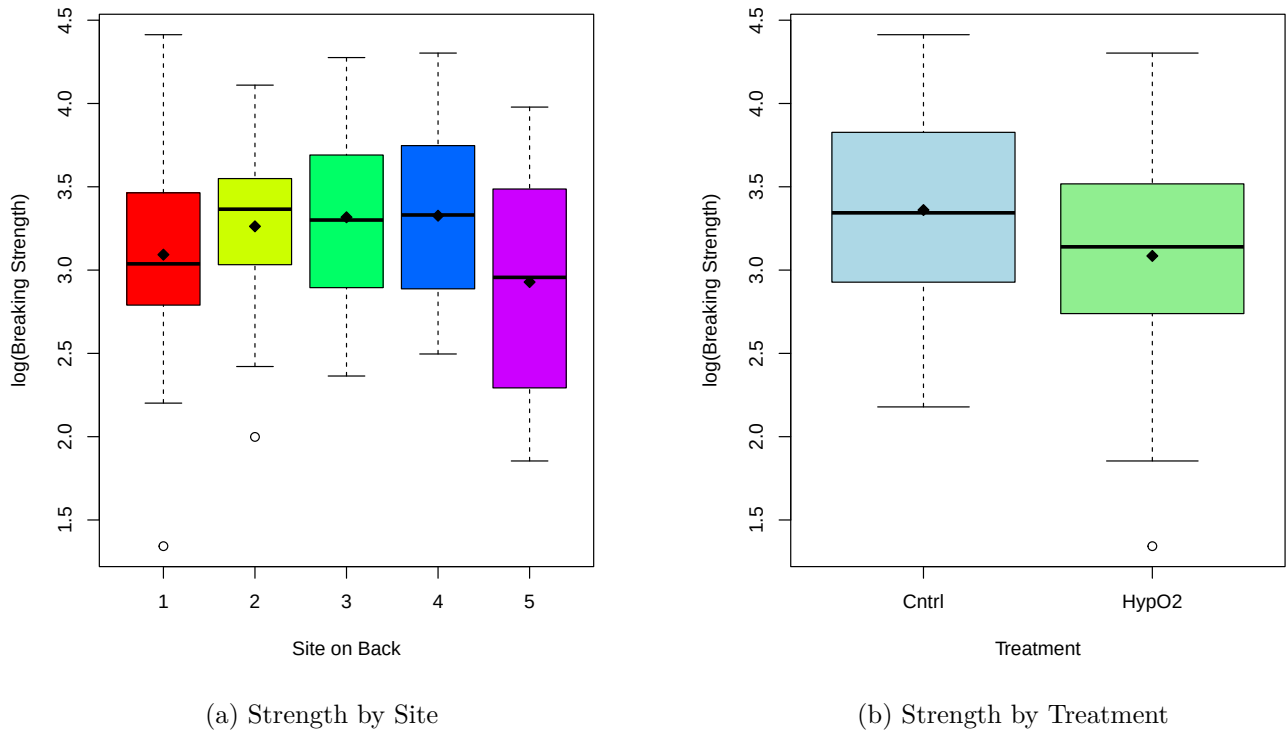


Figure 2: Visualisations of rat data

Task i)

An initial linear mixed model, Equation 3, is fitted with interactions between Treatment and Site.

$$\log Y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + d_k + \epsilon_{ijk} \quad (3)$$

Where:

- $\log Y_{ijk}$ is the log-transformed breaking energy for the k -th rat at the i -th site under the j -th treatment.
- μ is the overall average $\log Y$.

- α_i is the fixed main effect of the i -th site (1 through 5).
- β_j is the fixed main effect of the j -th treatment (Hyperbaric O2 or Control).
- $(\alpha\beta)_{ij}$ is the fixed interaction effect between the i -th site and the j -th treatment, representing how the treatment effect might change depending on the specific site.
- d_k is the random effect of the k -th rat, assumed to be normally distributed with a mean of zero and a variance of σ_{rat}^2 ($d_k \sim N(0, \sigma_{rat}^2)$).
- ϵ_{ijk} is the residual measurement error, assumed to be normally distributed with a mean of zero and a variance of σ^2 ($\epsilon_{ijk} \sim N(0, \sigma^2)$).

Running an F-test (`drop1`) shows a p-value value of 0.4292 for the interaction $(\alpha\beta)_{ij}$, i.e., the interaction is insignificant indicating it can be dropped.

A final linear mixed model, Equation 4, is fitted, omitting the interaction between Treatment and Site and keeping rat as a random effect.

$$\log Y_{ijk} = \mu + \alpha_i + \beta_j + d_k + \epsilon_{ijk} \quad (4)$$

Analysis of variance (ANOVA) of the final model showed that *site* was statistically significant with $p = 0.0087$ and is likely to effect the logarithm of the response Y , unlike *treatment* with $p = 0.1254$, see Table 1.

Table 1: Analysis of variance (ANOVA) of Equation 4.

	Sum Sq	Mean Sq	NumDF	DenDF	F value	p-value
site	3.0921390	0.7730347	4	77.1598	3.663568	0.0087355
treat	0.5359382	0.5359382	1	21.7507	2.539920	0.1254297

The variance from rat to rat is 0.1476 and the variance of the residual measurement is 0.211. The effect of each site can be seen in Table 2, relative to the intercept which is site 1. Sites 2,3,4 increase the logarithm of the response where site 5 decreases the logarithm of the response. The treatment which is deemed insignificant also decreases the logarithmic response.

Table 2: Fixed effects coefficients of Equation 4.

	Estimate	Std. Error	df	t value	p-value
(Intercept)	3.3069968	0.1658947	40.19088	19.934308	0.0000000
site2	0.1578629	0.1397460	76.71863	1.129642	0.2621468
site3	0.1814219	0.1361347	76.11829	1.332665	0.1866173
site4	0.2604877	0.1463306	77.37442	1.780131	0.0789804
site5	-0.2714586	0.1517373	78.18226	-1.789003	0.0774874
treatHypO2	-0.2937362	0.1843094	21.75070	-1.593713	0.1254297

Task ii)

Site 5 exhibited the lowest estimated marginal mean, as seen in Figure 3.

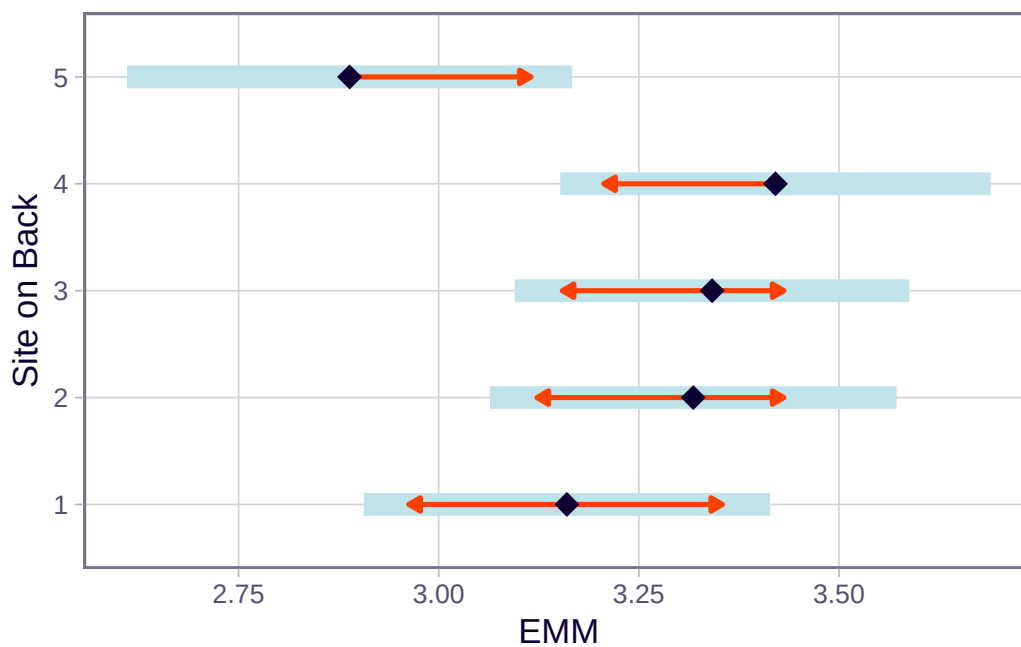


Figure 3: Estimated Marginal Mean of $\log Y$

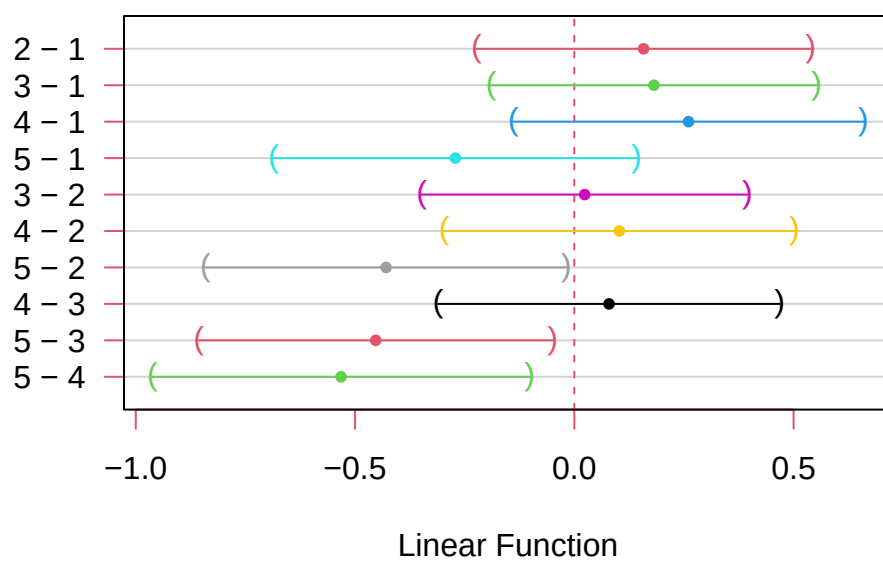


Figure 4: 95% family-wise confidence level: Site Differences

As can be seen in Figure 4, the pairwise comparisons involving site 5 showed that the 95% confidence intervals for the comparisons between site 5 and sites 2, 3, and 4 did not include zero, indicating significant differences. In contrast, the confidence interval for the comparison between sites 5 and 1 included zero, indicating no statistically significant difference between these two sites.

Task iii)

Fitting a linear model (Equation 5) suggest the *treatment* is significant due to the linear model underestimating the “true” variance which the LMM takes into account as can be seen in Table 3.

$$\log(Y) = \mu + \text{site} + \text{treat} + \varepsilon \quad (5)$$

None of the p-values indicate any statistical significance in Table 3.

Table 3: Coefficients of a fitted linear model.

	Estimate	Std. Error	t value	p-value
(Intercept)	3.2685638	0.1443489	22.6435040	0.0000000
site2	0.1699548	0.1783837	0.9527481	0.3430618
site3	0.2221808	0.1746302	1.2722931	0.2062795
site4	0.2615423	0.1856226	1.4090007	0.1620001
site5	-0.1828242	0.1911900	-0.9562432	0.3413024

F-test of the linear model, Equation 5, shows that reducing the model by removing *site* is possible ($p = 0.134$) and that *treatment* is significant ($p = 0.0138$).

Conversely, as shown previously in the analysis of variance of the LMM model, Table 1, *site* is a statistically significant variable ($p = 0.00874$) and *treatment* is not significant ($p = 0.125$), thus showing the opposite result of Equation 5.

Exercise 3

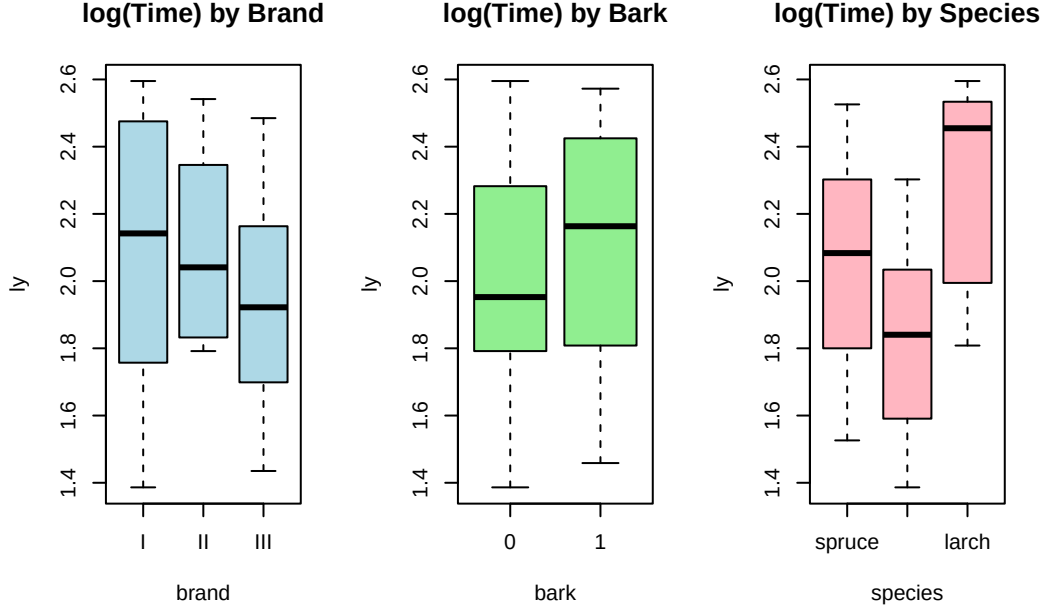


Figure 5: Exploratory plots of the “Saw” data.

Model:

$$\log(y_{ijklm}) = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \gamma_k + t_l + s_m + \epsilon_{ijklm}$$

Where:

- $\log(y_{ijklm})$ is the log-transformed cutting time for the i -th brand, j -th bark status, k -th species, used by the l -th team, with the m -th specific physical saw.
- μ is the overall average log-transformed cutting time.
- α_i is the fixed main effect of the i -th saw brand.
- β_j is the fixed main effect of the j -th bark status (e.g., bark vs. no bark).
- $(\alpha\beta)_{ij}$ is the fixed interaction effect between the i -th saw brand and the j -th bark status, representing whether the time saved by debarking depends on which brand of saw is used.
- γ_k is the fixed main effect of the k -th wood species (e.g., spruce, pine).
- t_l is the random effect of the l -th team, assumed to be normally distributed with a mean of zero and a variance of σ_{team}^2 ($t_l \sim N(0, \sigma_{team}^2)$).

- s_m is the random effect of the m -th physical saw, assumed to be normally distributed with a mean of zero and a variance of σ_{saw}^2 ($s_m \sim N(0, \sigma_{saw}^2)$).
- ϵ_{ijklm} is the residual measurement error, assumed to be normally distributed with a mean of zero and a variance of σ^2 ($\epsilon_{ijklm} \sim N(0, \sigma^2)$).

Fitting this model shows the interaction is not necessary with a p-value of 0.8730176 on the interaction when performing a F-test

Final Model:

$$\log(y_{ijklm}) = \mu + \alpha_i + \beta_j + \gamma_k + t_l + s_m + \epsilon_{ijklm}$$

Random effects:

Groups	Name	Variance	Std.Dev.
team	(Intercept)	0.079568	0.28208
saw_id	(Intercept)	0.001022	0.03198
Residual		0.008846	0.09405

Number of obs: 36, groups: team, 6; saw_id, 6

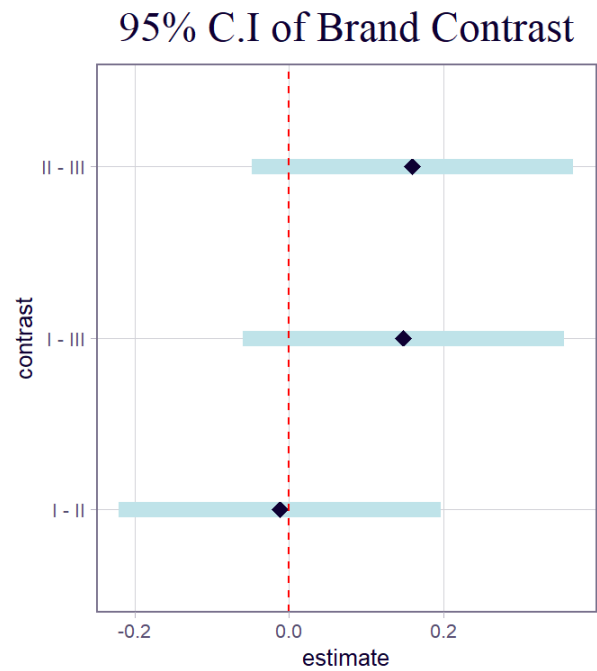
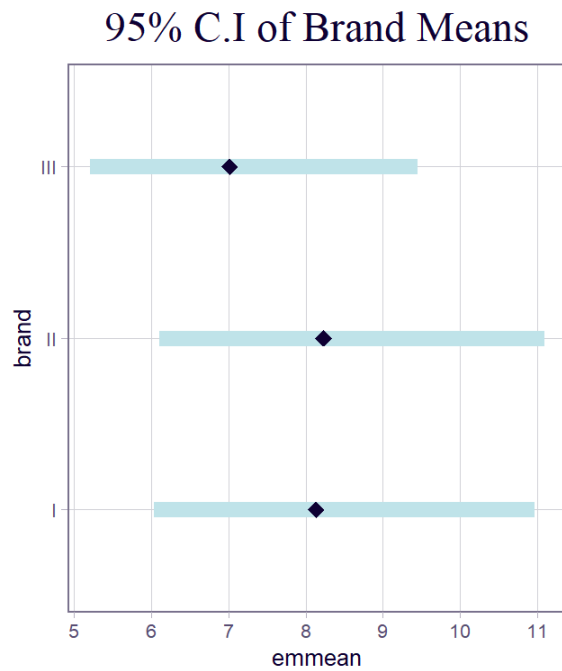
Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	2.00543	0.12348	6.27991	16.241	2.29e-06 ***
brandII	0.01176	0.04997	2.99998	0.235	0.8291
brandIII	-0.14837	0.04997	2.99998	-2.969	0.0591 .
speciespine	-0.21274	0.03840	22.00000	-5.540	1.44e-05 ***
specieslarch	0.25630	0.03840	22.00000	6.675	1.04e-06 ***
bark1	0.15228	0.03135	22.00000	4.857	7.45e-05 ***

Findings:

Fixed Effects : Brand appears to not be significantly significant with the others being significant.

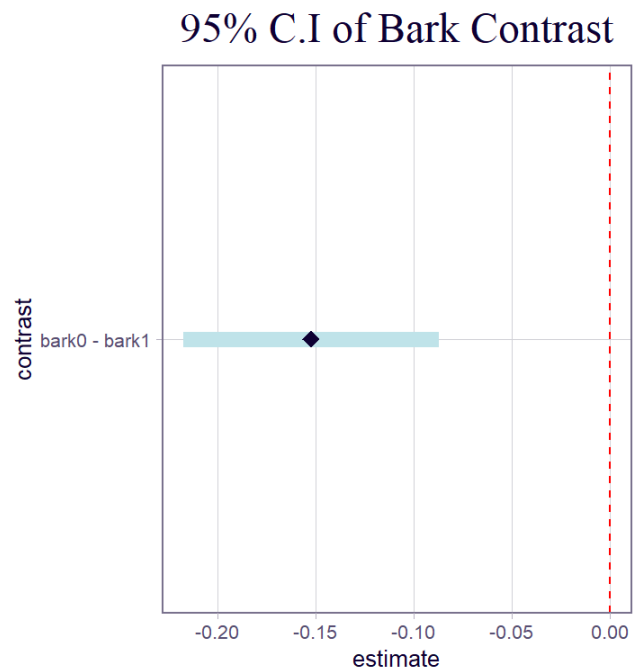
Random Effects : Saw_id has very small variance indicating it has very little effect on the final results. E.g using saw A or D makes very small differences.



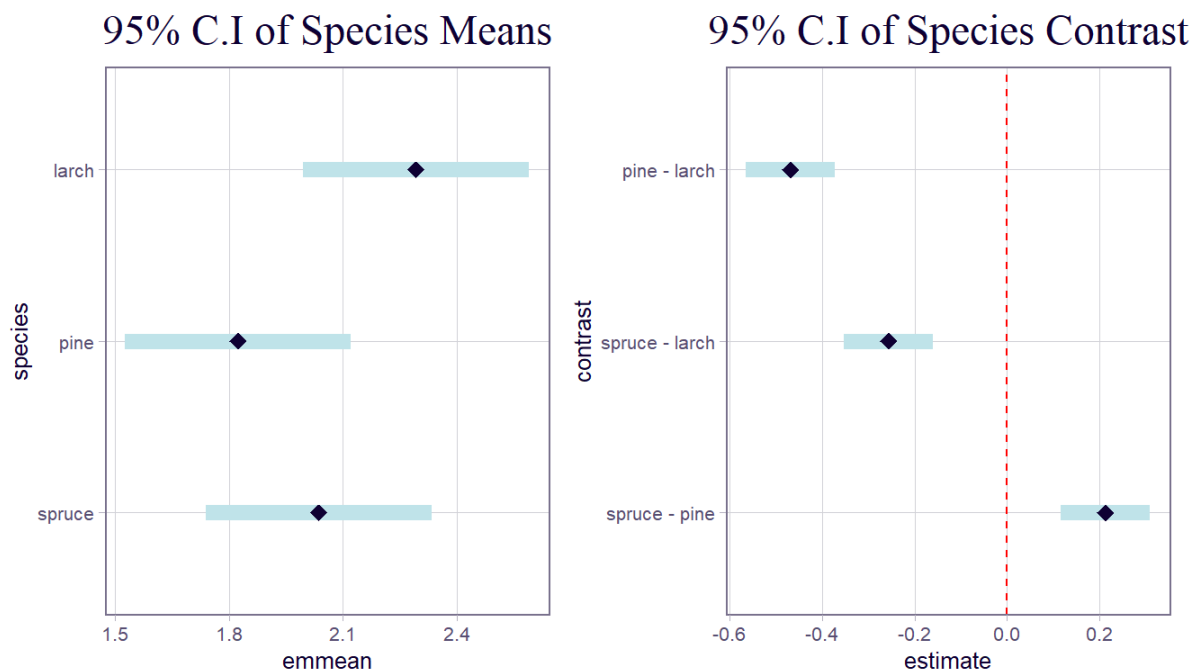
Part i)

As can be seen all brands perform fairly similar and there can be no statistical results in the difference of cutting time due to brand type. Confidence interval of all contrast contains 0.

Part ii)



It can be seen that bark is significantly significant reducing the cutting time by about 0.15 minutes with fairly high confidence not containg 0. Insert p-value



Part iii)

All species are statistically significant. Insert p-values